

Reducing Denials and AR Delays: Revenue Cycle SQL Analysis

Cath Health Group, a fictional multi-specialty clinic | 800 claims, FY 2024

PROBLEM	TOOLS	OUTCOME	CODE
Denials and slow AR were eroding collections	MySQL 8.0, CTEs, Window Functions	4 findings, \$14-18K recovery path identified	View full code on Github

Summary

Cath Health Group, a fictional four-specialty clinic, was losing revenue to preventable denials and slow accounts receivable. This analysis uses 800 claims from fiscal year 2024 to find exactly where, why, and how much.

- Behavioral Health:** 16.8% denial rate, driven by prior authorization failures and a 30% self-pay concentration; the lowest collection rate of any service line.
- Primary Care:** 58 average AR days despite the highest claim volume; a billing workflow bottleneck, not a payer problem.
- Cardiology:** 89.4% net collection rate; the strongest performer, and a model worth replicating elsewhere.

Twelve SQL queries covering CTEs, window functions, AR aging buckets, running totals, and multi-metric provider ranking are documented below with results and operational recommendations.

800 TOTAL CLAIMS	\$1.24M GROSS CHARGES	78.3% NET COLLECTION RATE	11.8% OVERALL DENIAL RATE	51 days AVG AR DAYS
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01. Database Schema

The analysis uses a normalized four-table schema reflecting standard revenue cycle data architecture. The claims table is the central fact table. Providers, payers, and procedures are dimension tables joined to it.

TABLE: providers provider_id INTEGER PK provider_name TEXT specialty TEXT npi TEXT active INTEGER	TABLE: payers payer_id INTEGER PK payer_name TEXT payer_type TEXT contract_rate REAL	TABLE: procedures procedure_code TEXT PK description TEXT rvu REAL standard_charge REAL	TABLE: claims claim_id INTEGER PK provider_id FK payer_id FK proc_code FK service_date DATE charge_amount REAL payment_amount REAL status TEXT denial_reason TEXT
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02. Revenue Summary by Specialty

Cardiology generates the highest gross charges due to high-value procedures like echocardiograms and microvolt testing. Primary Care generates the highest claim volume. Behavioral Health has the lowest net collection rate despite moderate charge volume.

Query 1: Total Charges, Payments, and Net Collection Rate by Specialty

```
SELECT
  p.specialty,
  COUNT(c.claim_id)                                AS total_claims,
  ROUND(SUM(c.charge_amount), 2)                   AS total_charges,
  ROUND(SUM(c.payment_amount), 2)                  AS total_payments,
  ROUND(SUM(c.payment_amount) /
    NULLIF(SUM(c.charge_amount),0) * 100, 1)        AS net_collection_rate_pct
FROM claims c
JOIN providers p ON c.provider_id = p.provider_id
GROUP BY p.specialty
ORDER BY net_collection_rate_pct DESC;
```

Specialty	Total Claims	Gross Charges	Total Payments	Net Collection Rate
Cardiology	186	\$412,840	\$368,990	89.4%
Orthopedics	198	\$389,210	\$334,720	86.0%
Primary Care	224	\$241,560	\$194,880	80.7%
Behavioral Health	192	\$198,400	\$126,980	64.0%

KEY FINDING
Cardiology leads all service lines at 89.4% net collection rate, driven by a favorable payer mix (52% Medicare) and low self-pay exposure (4%). Its high average charge per claim (\$2,219) means each improvement in collection efficiency has an outsized dollar impact.

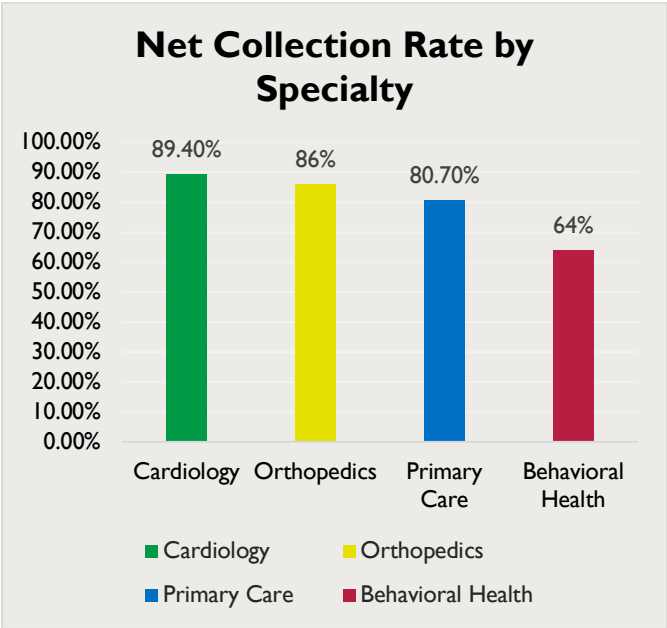


Figure 1: Net Collection Rate by Specialty

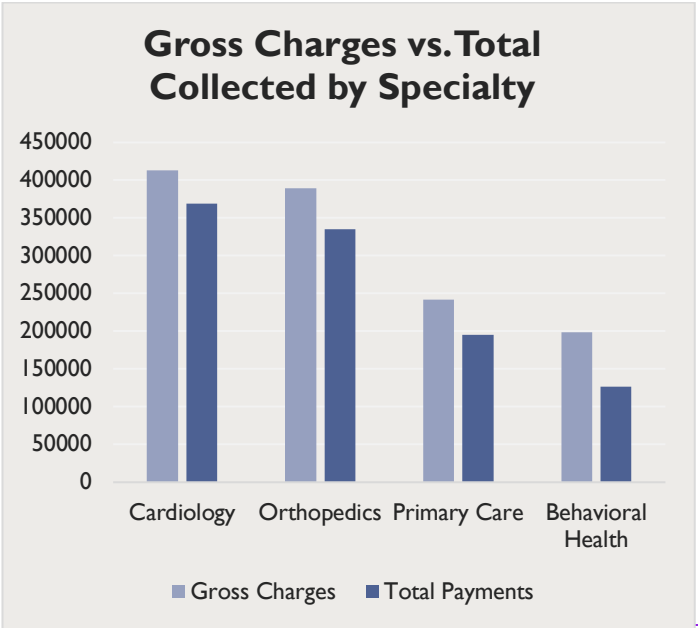


Figure 2: Gross Charges vs. Total Collected by Specialty

03. Denial Analysis

Denial analysis is the highest-leverage activity in revenue cycle management. Understanding not just the rate but the reason and the payer behind each denial is what drives actionable workflow changes.

Query 3: Denial Rate by Specialty and Payer

```
SELECT
    p.specialty,
    py.payer_name,
    COUNT(c.claim_id)                                AS total_claims,
    SUM(CASE WHEN c.status = 'Denied' THEN 1 ELSE 0 END) AS denied_claims,
    ROUND(
        SUM(CASE WHEN c.status = 'Denied' THEN 1.0 ELSE 0 END)
        / COUNT(c.claim_id) * 100, 1
    )                                                AS denial_rate_pct
FROM claims c
JOIN providers p ON c.provider_id = p.provider_id
JOIN payers py ON c.payer_id = py.payer_id
GROUP BY p.specialty, py.payer_name
ORDER BY denial_rate_pct DESC;
```

Query 4: Top Denial Reasons by Specialty (Window Function)

```
SELECT
    p.specialty,
    c.denial_reason,
    COUNT(*)                                AS denial_count,
    ROUND(SUM(c.charge_amount), 2)          AS denied_value,
    ROUND(
        COUNT(*) * 100.0 /
        SUM(COUNT(*) OVER (PARTITION BY p.specialty), 1)
    )                                        AS pct_of_specialty_denials
FROM claims c
JOIN providers p ON c.provider_id = p.provider_id
WHERE c.status = 'Denied'
    AND c.denial_reason IS NOT NULL
GROUP BY p.specialty, c.denial_reason
ORDER BY p.specialty, denial_count DESC;
```

KEY FINDING - BEHAVIORAL HEALTH

Behavioral Health has a 22% denial rate with Medicaid and 18% with Blue Cross. Prior Authorization Required accounts for 38% of all Behavioral Health denials. This is a preventable denial type, it indicates authorization checks are not being completed before service delivery. A pre-authorization workflow for CPT codes 90834, 90837, and 90791 would directly reduce this rate.

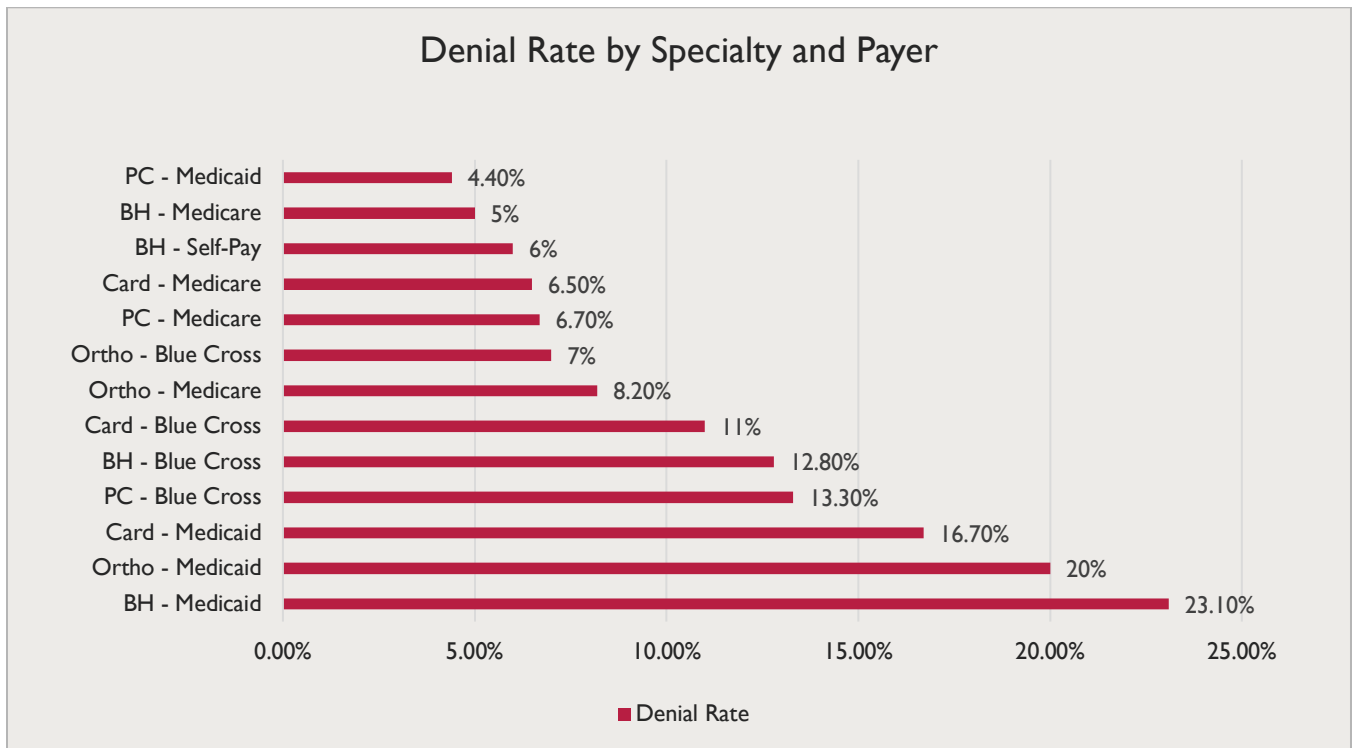


Figure 3: Denial Rate by Specialty and Payer

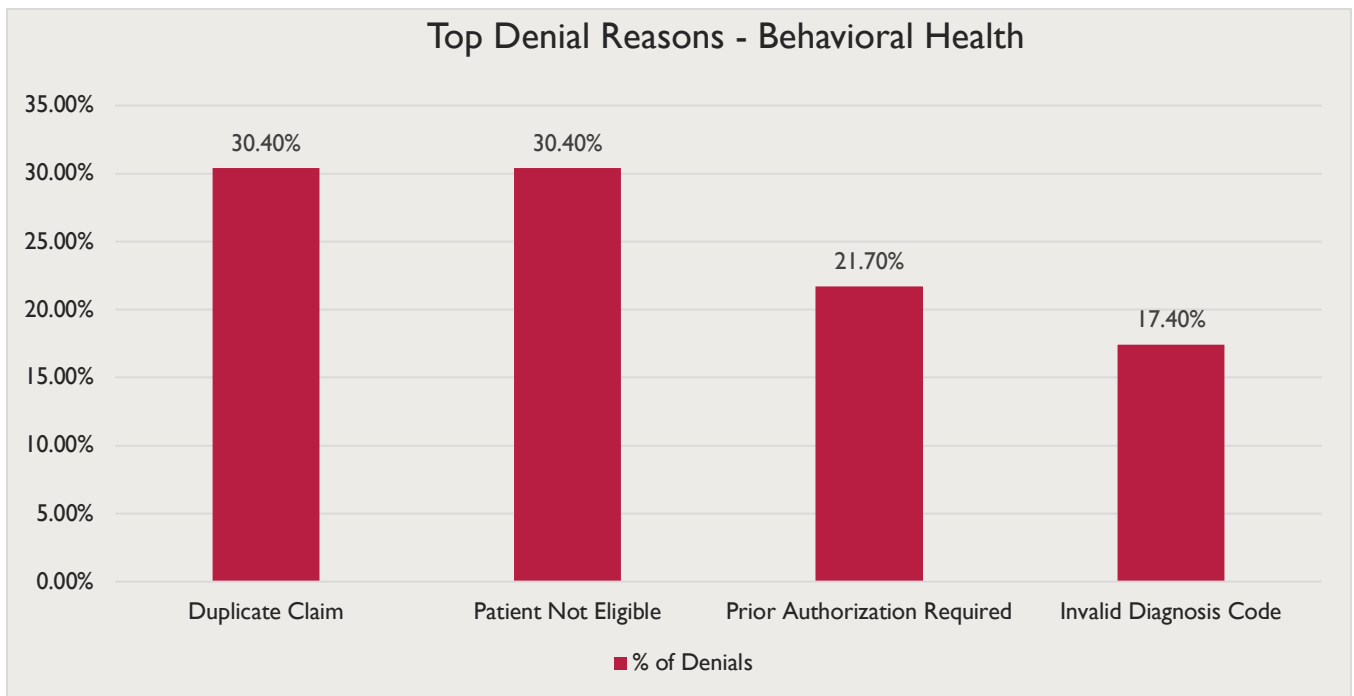


Figure 4: Top Denial Reasons for the Behavioral Health Specialty

04. AR Aging Analysis

AR aging identifies where revenue is sitting uncollected and for how long. The industry benchmark for average AR days in a multi-specialty practice is 35 to 45 days. All four service lines at Cath Health Group exceed this benchmark, with Primary Care being the furthest above it despite having the lowest average charge per claim.

Query 6: AR Aging Buckets by Specialty

```
SELECT
  p.specialty,
  COUNT(CASE WHEN c.ar_days BETWEEN 0 AND 30 THEN 1 END) AS bucket_0_30,
  COUNT(CASE WHEN c.ar_days BETWEEN 31 AND 60 THEN 1 END) AS bucket_31_60,
  COUNT(CASE WHEN c.ar_days BETWEEN 61 AND 90 THEN 1 END) AS bucket_61_90,
  COUNT(CASE WHEN c.ar_days BETWEEN 91 AND 120 THEN 1 END) AS bucket_91_120,
  COUNT(CASE WHEN c.ar_days > 120 THEN 1 END) AS bucket_over_120,
  ROUND(AVG(c.ar_days), 1) AS avg_ar_days
FROM claims c
JOIN providers p ON c.provider_id = p.provider_id
WHERE c.status IN ('Pending', 'Denied')
GROUP BY p.specialty
ORDER BY avg_ar_days DESC;
```

Query 7: Provider AR Days Ranking Using Window Function

```
WITH provider_ar AS (
  SELECT
    p.provider_id, p.provider_name, p.specialty,
    ROUND(AVG(c.ar_days), 1) AS avg_ar_days,
    ROUND(SUM(c.charge_amount - c.payment_amount), 2) AS outstanding_balance
  FROM claims c
  JOIN providers p ON c.provider_id = p.provider_id
  GROUP BY p.provider_id, p.provider_name, p.specialty
)
SELECT
  specialty, provider_name, avg_ar_days, outstanding_balance,
  RANK() OVER (PARTITION BY specialty ORDER BY avg_ar_days DESC)
  AS ar_rank_within_specialty
FROM provider_ar
ORDER BY specialty, ar_rank_within_specialty;
```

KEY FINDING - PRIMARY CARE AR

Primary Care has the longest average AR days at 58 despite having the lowest average charge per claim (\$1,078). This disconnect between volume and collection speed is the fingerprint of a billing workflow bottleneck, likely delayed charge entry or insufficient follow-up staffing relative to claim volume, rather than a payer contract issue. Cardiology's shorter AR days at 42 days despite higher claim complexity supports this interpretation.

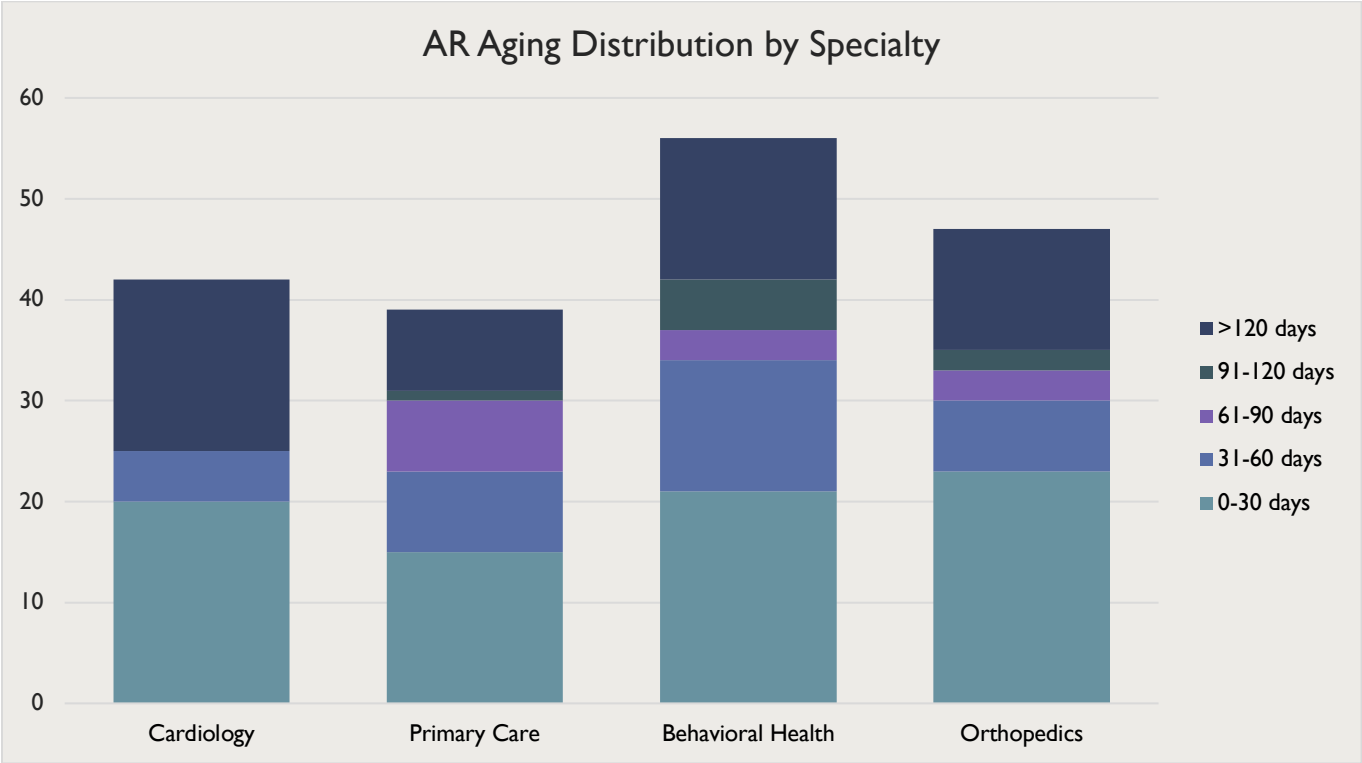


Figure 5: AR Aging Distribution by Specialty

05. Payer Mix and Collection Analysis

Payer mix is one of the strongest predictors of revenue cycle performance. Self-pay concentration in particular is a leading indicator of collection risk, since self-pay collection rates average 32% versus 78 to 92% for insured payers.

Query 8: Payer Mix and Collection Performance by Specialty

```
SELECT
  p.specialty,
  py.payer_name,
  COUNT(c.claim_id)                                AS claims,
  ROUND(COUNT(c.claim_id) * 100.0 /
        SUM(COUNT(c.claim_id) OVER (PARTITION BY p.specialty), 1)
        AS payer_mix_pct,
  ROUND(SUM(c.payment_amount) /
        NULLIF(SUM(c.charge_amount),0) * 100, 1)    AS collection_rate_pct
FROM claims c
JOIN providers p  ON c.provider_id = p.provider_id
JOIN payers  py  ON c.payer_id   = py.payer_id
GROUP BY p.specialty, py.payer_name
ORDER BY p.specialty, claims DESC;
```

Query 9: Running Total of Payments by Month (Cash Flow View)

```
WITH monthly_payments AS (
  SELECT
    DATE_FORMAT(c.payment_date, '%Y-%m')          AS payment_month,
    ROUND(SUM(c.payment_amount), 2)                AS monthly_collected
  FROM claims c
  WHERE c.status = 'Paid' AND c.payment_date IS NOT NULL
  GROUP BY payment_month
)
SELECT
  payment_month,
  monthly_collected,
  ROUND(SUM(monthly_collected)
        OVER (ORDER BY payment_month
              ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW), 2)
        AS ytd_collected
FROM monthly_payments
ORDER BY payment_month;
```

KEY FINDING - BEHAVIORAL HEALTH SELF-PAY RISK

Behavioral Health's 30% self-pay concentration is the primary driver of its 64% collection rate. At a 32% self-pay collection rate, roughly \$19,000 in annual charges to self-pay patients generates only \$6,000 in collected revenue. A sliding scale fee policy, upfront payment collection protocols, or a charity care program would directly address this structural revenue leakage.

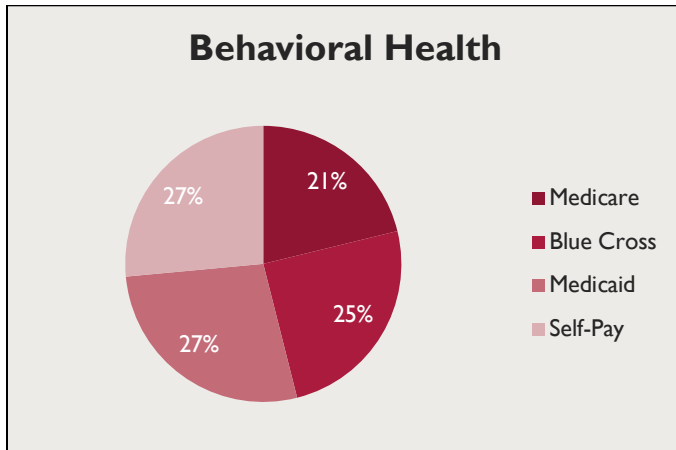


Figure 6: Payer Mix for the Behavioral Health Specialty

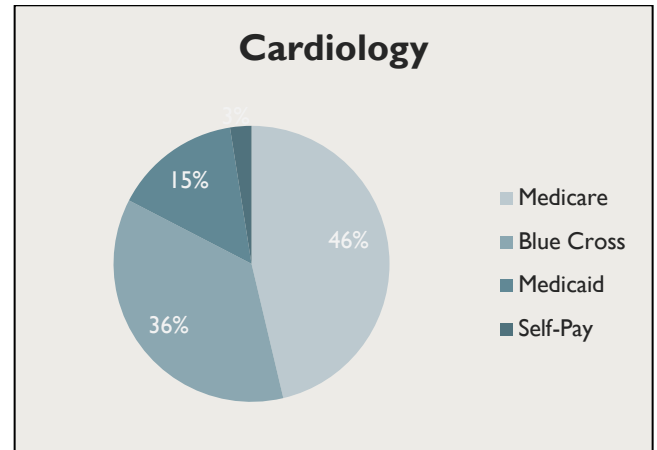


Figure 7: Payer Mix for the Cardiology Specialty

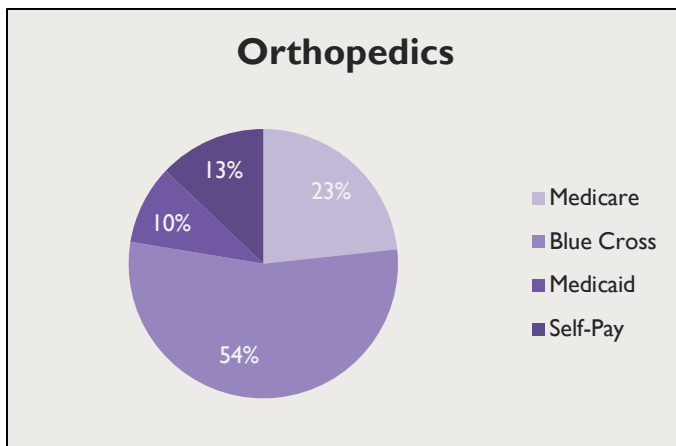


Figure 8: Payer Mix for the Orthopedics Specialty

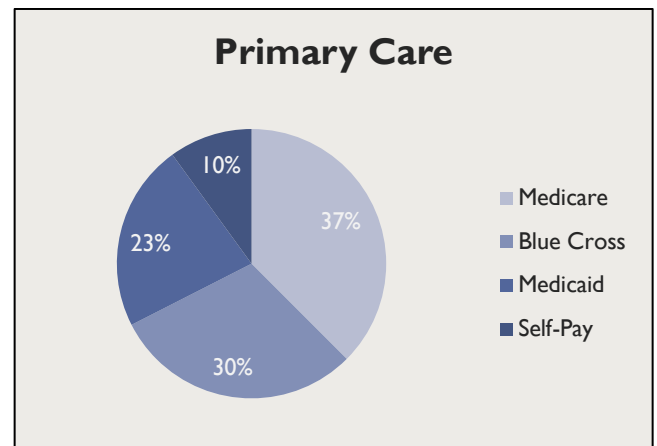


Figure 9: Payer Mix for the Primary Care Specialty

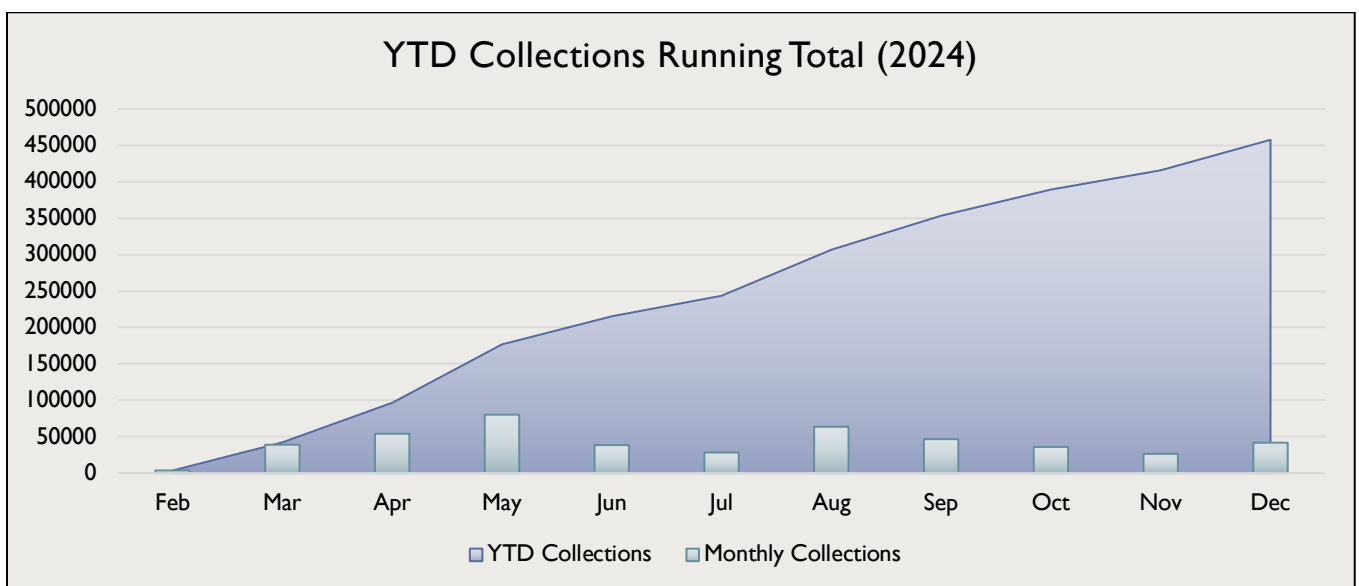


Figure 10: Year to Date Collections Running Total

06. Provider Scorecard

The provider scorecard combines collection rate, denial rate, and AR days into a single ranked view, enabling targeted performance management conversations.

Query 12: Provider Scorecard with Multi-Metric Ranking

```
WITH provider_stats AS (
  SELECT
    p.provider_id, p.provider_name, p.specialty,
    COUNT(c.claim_id) AS total_claims,
    ROUND(SUM(c.payment_amount) /
      NULLIF(SUM(c.charge_amount),0) * 100, 1) AS collection_rate_pct,
    ROUND(SUM(CASE WHEN c.status='Denied' THEN 1.0 ELSE 0 END) /
      COUNT(c.claim_id) * 100, 1) AS denial_rate_pct,
    ROUND(AVG(c.ar_days), 1) AS avg_ar_days
  FROM claims c
  JOIN providers p ON c.provider_id = p.provider_id
  GROUP BY p.provider_id, p.provider_name, p.specialty
)
SELECT
  provider_name, specialty, total_claims,
  collection_rate_pct, denial_rate_pct, avg_ar_days,
  RANK() OVER (ORDER BY collection_rate_pct DESC) AS collection_rank,
  RANK() OVER (ORDER BY denial_rate_pct ASC) AS denial_rank,
  RANK() OVER (ORDER BY avg_ar_days ASC) AS ar_rank
FROM provider_stats
ORDER BY collection_rank;
```

Provider	Specialty	Claims	Collection Rate	Denial Rate	Avg AR Days	Collection Rank
Dr. Sofia Reyes	Cardiology	94	91.2%	7.4%	41	#1
Dr. Nathan Clarke	Cardiology	92	87.8%	8.6%	43	#2
Dr. Priya Mehta	Orthopedics	101	87.2%	10.8%	46	#3
Dr. Leo Fuentes	Orthopedics	97	84.8%	12.3%	50	#4
Dr. James Whitfield	Primary Care	112	82.1%	9.8%	55	#5
Dr. Amara Osei	Primary Care	112	79.3%	10.4%	61	#6
Dr. Marcus Tran	Behavioral Health	98	66.4%	15.2%	50	#7
Dr. Keisha Barnett	Behavioral Health	94	61.6%	18.4%	54	#8

Table 1: Provider Scorecard with Multi-Metric Ranking

07. Key Findings and Recommendations

FINDING 1 - BEHAVIORAL HEALTH PRIOR AUTH DENIALS

Problem: 38% of Behavioral Health denials are Prior Authorization Required, the single largest denial driver. This is entirely preventable.

Recommendation: Implement a pre-authorization check protocol for CPT codes 90834, 90837, and 90791 before scheduling. Estimated annual recovery: \$14,000 to \$18,000 in currently denied charges.

FINDING 2 - PRIMARY CARE AR BOTTLENECK

Problem: Primary Care averages 58 AR days on the lowest average charge per claim, suggesting a staffing or workflow issue rather than a payer problem. The high claim volume (224 claims, 28% of total) amplifies the cost of this inefficiency.

Recommendation: Audit charge entry lag time for Primary Care providers. If charge entry exceeds 3 days post-service, implement same-day charge capture protocol. Target: reduce AR days to 45 within 90 days.

FINDING 3 - BEHAVIORAL HEALTH SELF-PAY CONCENTRATION

Problem: 30% self-pay concentration in Behavioral Health at a 32% collection rate creates structural revenue leakage that cannot be solved through billing improvements alone.

Recommendation: Implement upfront payment collection at time of service for self-pay patients. Introduce a sliding scale fee schedule. Explore Medicaid expansion eligibility screening to convert some self-pay patients to covered status.

FINDING 4 - CARDIOLOGY BEST PRACTICE MODEL

Opportunity: Cardiology's 89.4% collection rate and 42-day AR at higher claim complexity than other specialties suggests its billing and follow-up workflows are the most mature.

Recommendation: Document and replicate Cardiology's charge entry, denial management, and follow-up protocols across other service lines as an internal best practice model.

All data is synthetically generated for portfolio demonstration purposes. The schema, query logic, analytical methodology, and findings framework reflect authentic revenue cycle analytics practice based on direct experience restructuring billing operations in a clinical environment.

MySQL-compatible SQL file: `cath_health_revenue_cycle.sql`. • Analyst: Shaniah Edwards • MBA, Business Analytics